

Achuthanand Madanapalli

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SUMMARY

CMU Master's student in AI Engineering (Information Security), graduating May 2028, with a BS in Computer Science from George Mason University (Honors College). Full-stack (React, Next.js, TypeScript) and applied ML (PyTorch, scikit-learn) experience from internships at Envisn Inc. and Excelerate and projects including an 89%-accuracy recommendation engine. Seeking a Summer 2027 AI, software engineering, or security internship to build production-scale, reliable and ethical ML-driven systems.

EDUCATION

Carnegie Mellon University (CMU) – Pittsburgh, PA

Graduation: May 2028

MS in Artificial Intelligence Engineering – Information Security

Relevant Coursework Experience: Systems & Toolchains for AI Engineers, Intro to ML with Adversaries in Mind, Intro to Info Sec

George Mason University (GMU) – Honors College, Fairfax, VA

Graduated: December 2025

Bachelor of Science: Computer Science (Major) and Minor in Business

GPA: 3.73

Relevant Coursework: Python, C Language, Java Data Structures, Marketing, Management, Research, Operating Systems, Software Engineering, Statistics, Relational Databases, Networking, Artificial Intelligence

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, C#, C++, SQL, HTML

Frameworks and Libraries: React, Next.js, Tailwind CSS, PyTorch, scikit-learn, Pandas, NumPy

Tools and Platforms: MongoDB, Unix, JUnit, Jest, Vitest, Agile, Google Cloud Platform, AWS

HONORS AND CERTIFICATIONS

- **AWS Cloud Practitioner Certification** - Learning how to navigate through Amazon Web Services.
- **Azure AI Fundamentals** - Learning how to navigate through Azure AI services.
- **Headstarter Fellowship** - Made projects that helped me learn about Next.js and React independently.
- **Inspirit Innovators AI Innovation Fellowship Certification** - ML Track: Logistic regression, CNN, scikit-learn, TensorFlow.

PROJECTS

- **Project 'Protein Classifier'**
 - Built a protein subcellular localization classifier using sequence-derived features extracted with BioPython.
 - Implemented and evaluated multiple machine learning models including Logistic Regression and SVM (linear and RBF kernels) using scikit-learn, comparing performance with Accuracy, F1-score, AUC, and confusion matrices.
 - Performed feature scaling, train/validation/test splitting, and model interpretation using coefficient-based feature importance, leveraging NumPy, pandas, and matplotlib/seaborn for data processing and visualization.
- **Project 'Recommendation Engine'**
 - Predicted drop-off and sign-up rates for courses the platform using Python libraries and datasets provided.
 - Created a recommendation engine using Logistic Regression, Decision Tree, Random Forest and XGBoost.
 - Project involved 89% accuracy during the making of the model with training and testing datasets
- **Project 'SpendFlow'**
 - AI expense tracker that uses Next.js and MongoDB to provide simplified budgeting and personalized financial insights
 - Secure authentication with Clerk, a responsive and intuitive UI, and AI-driven analytics to deliver spending recommendations
 - Used Tailwind CSS for design improvements with futuristic aesthetic
- **Project 'GradeMe'**
 - Collaborated with a team across 3 Agile sprints as team lead, a grade calculator enabling account creation and grade storage for multiple classes
 - Implemented core business logic and website functionality, ensuring seamless user experience and data accuracy
 - Utilized HTML, CSS, and JavaScript for dynamic front-end development, integrating Jest for rigorous unit testing to validate critical application features. PHP was used for backend and saving the changes of certain course grades

EXPERIENCE

- **Undergraduate Teaching Assistant - George Mason University** **August 2024 – December 2025**
 - Helped students of Software engineering and created JAR files and database integrated models for students to use in projects
 - Helped with CS 367 (Computer Systems and Programming) to advise and aid in questions and project-making
- **Software Developer Internship - Envisn Inc.** **August 2024 – August 2025**
 - Analyzed and improved current system integrations and migration strategies for the NetVisn product
 - Participated in all aspects of agile software development including design, implementation, and deployment
 - Explored ways to enhance data quality and reliability of the applications
- **AI-Powered Data Insights Internship - Excelerate** **May 2025 – July 2025**
 - Operated as project leader of a team to deliver assignments on time
 - Created final project and presentation that demonstrated improvements to the signup program and recommendations.